

Product Carbon Footprint Study Report

Product Name: cookstove, gas or electric

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: cookstove, gas or electric

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item cookstove

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item cookstove.

2. System Boundary

cookstove production, gas or electric, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of cookstove, gas or electric produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **151.487 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

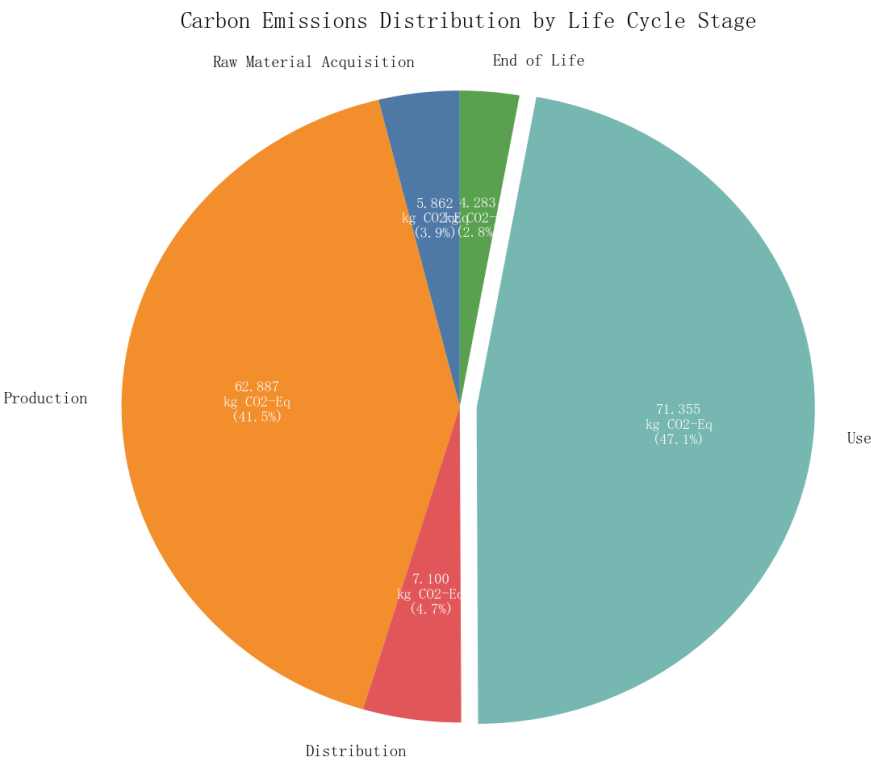


Table 1: Carbon Emissions by Life Cycle Stage

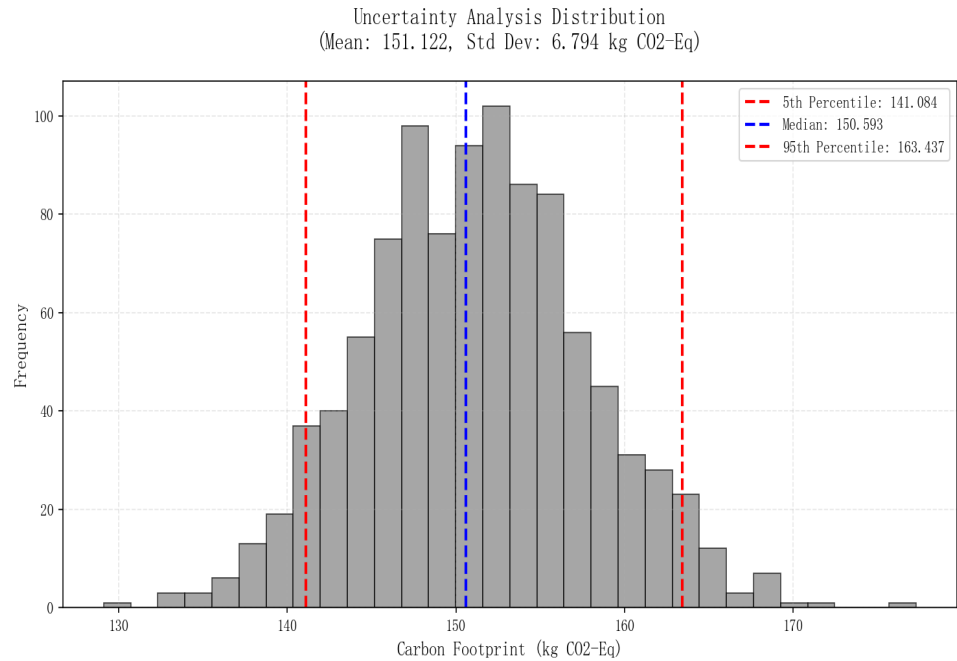
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	5.862	3.87
Production	62.887	41.51
Distribution	7.100	4.69

Use	71.355	47.10
End of Life	4.283	2.83
Total	151.487	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	151.122
Standard Deviation	6.794
Coefficient of Variation (CV)	4.50%
Median	150.593
5th Percentile	141.084
95th Percentile	163.437

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the cookstove, gas or electric, is 151.487 kg CO₂-Eq over its full life cycle. This value represents a moderate emission intensity for a household appliance, reflecting combined impacts from material sourcing through disposal. The assessment follows a cradle-to-grave approach, capturing all relevant stages under standard LCA methodology.

Lifecycle contributions reveal two dominant stages: the use phase accounts for 71.355 kg CO₂-Eq (47.1%), driven primarily by energy consumption during operation, while the production phase contributes 62.887 kg CO₂-Eq (41.5%), linked to manufacturing processes and component fabrication. Remaining stages—raw materials (3.9%), distribution (4.7%), and end-of-life (2.8%)—collectively represent only 11.4% of total emissions, indicating that operational and manufacturing efficiencies offer the greatest leverage for reduction.

Major emission sources are therefore concentrated in the use and production stages. The use phase's high share stems from electricity or gas consumption patterns, whereas production impacts arise from energy-intensive assembly and material processing. Targeted interventions in these areas would yield the most substantial footprint reductions.

2. Statement of Assumptions and Limitations

Key assumptions in this study include average European or global grid electricity mixes for the use phase, standard product lifetimes of 10–15 years, and representative operational profiles based on typical household cooking patterns. Transportation distances for distribution were modeled using default logistics scenarios, and end-of-life treatment assumed conventional recycling and disposal rates without region-specific waste management variations.

Limitations arise from reliance on secondary data sources such as generic life-cycle inventory databases, which may not fully capture proprietary manufacturing processes or local supply-chain specifics. Data quality for the production stage could be affected by aggregated industrial averages, potentially under- or over-estimating emissions for specialized components. Regional electricity grid decarbonization trends or user behavior variability were not dynamically modeled.

System boundaries are restricted to a cradle-to-grave scope encompassing raw material extraction, production, distribution, use, and end-of-life. Excluded elements include upstream infrastructure impacts, indirect land-use changes, and any ancillary consumer behaviors such as maintenance or accessory production. The assessment focuses exclusively on climate change impacts and does not extend to other environmental categories.

3. Suggestions for Improvement

- Optimize use-phase efficiency by integrating smart controls, induction technology, or higher-efficiency burners to reduce energy consumption and lower the 47.1% use-stage emissions.
- Shift production toward low-carbon manufacturing processes, such as renewable-powered facilities or lightweight material substitution, targeting the 41.5% production contribution.

- Engage suppliers to adopt recycled metals and low-impact components in raw material sourcing, indirectly supporting both production and end-of-life improvements.
- Implement product take-back programs and design-for-disassembly features to enhance recycling rates and reduce end-of-life burdens beyond the current 2.8%.
- Collect primary data from actual manufacturing sites and conduct sensitivity analyses on use patterns to improve overall data quality and reduce reliance on generic database values.